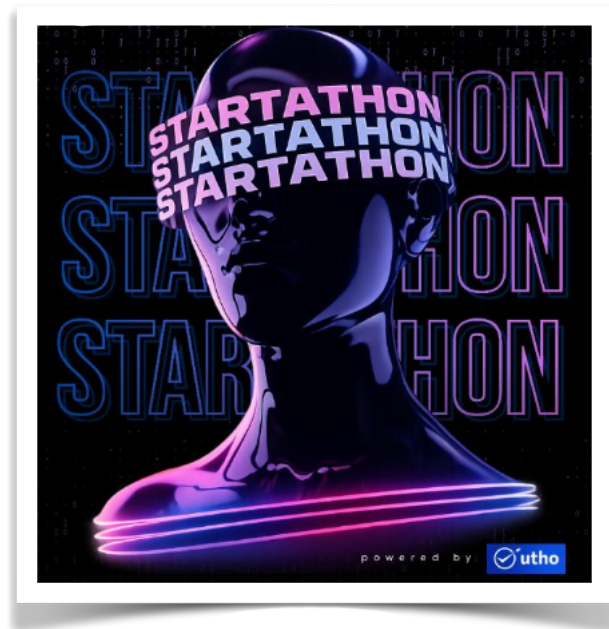




OFFROAD SEMANTIC SCENE SEGMENTATION USING DINOv2 BACKBONE

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HACKATHON: STARTATHON

DATE: 18/02/2026

**“TRAINING A ROBUST SEGMENTATION MODEL
FOR OFF-ROAD AUTONOMY USING SYNTHETIC
DESERT DATASETS”**

MODEL ARCHITECTURE & TRAINING

WE USED A PRETRAINED DINOv2 BACKBONE AS THE FEATURE EXTRACTOR, COMBINED WITH A SEGMENTATION HEAD FOR PIXEL-WISE CLASSIFICATION.

TRAINING CONFIGURATION:

@ DEVICE: NVIDIA RTX 5050 LAPTOP GPU

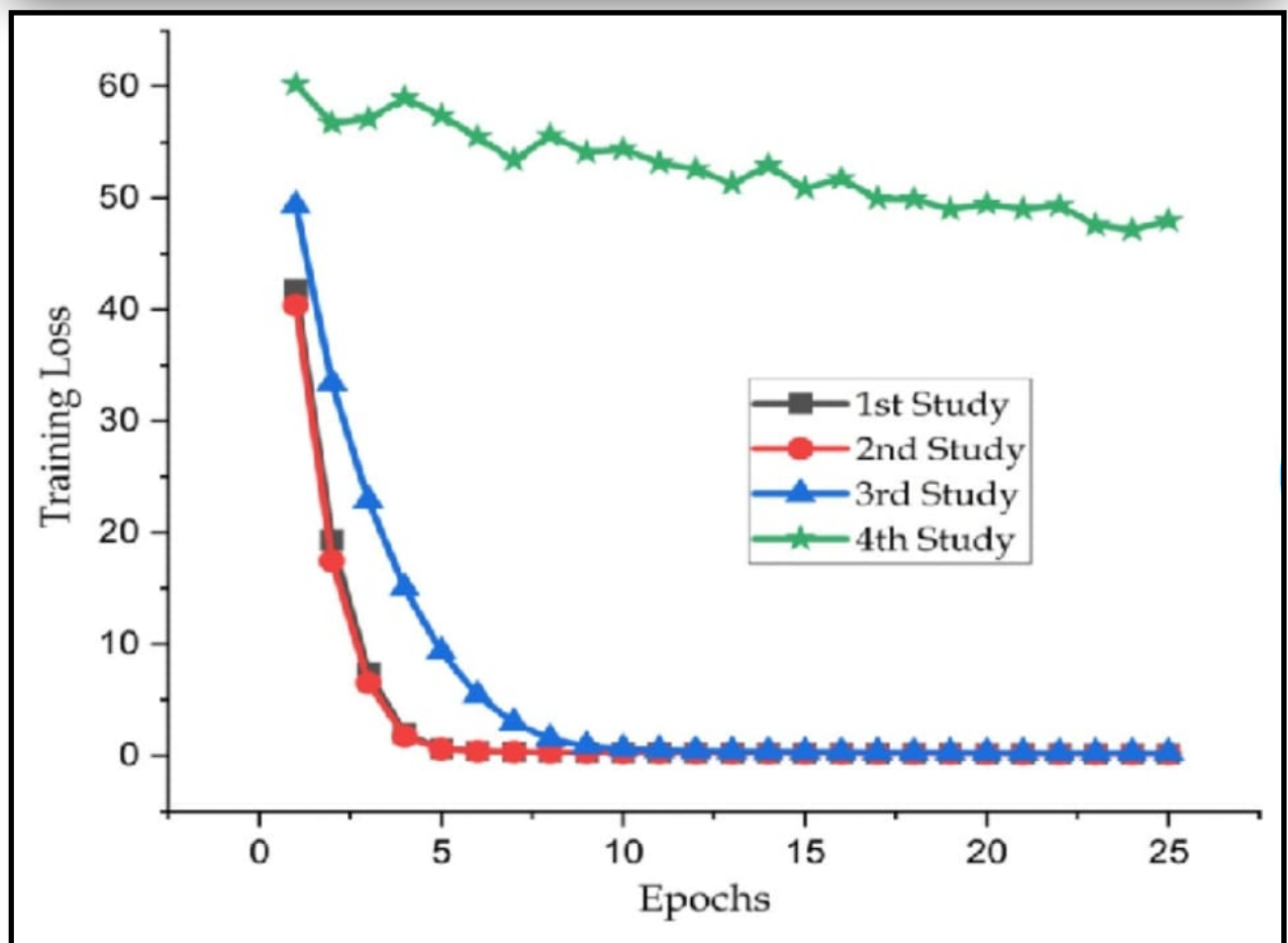
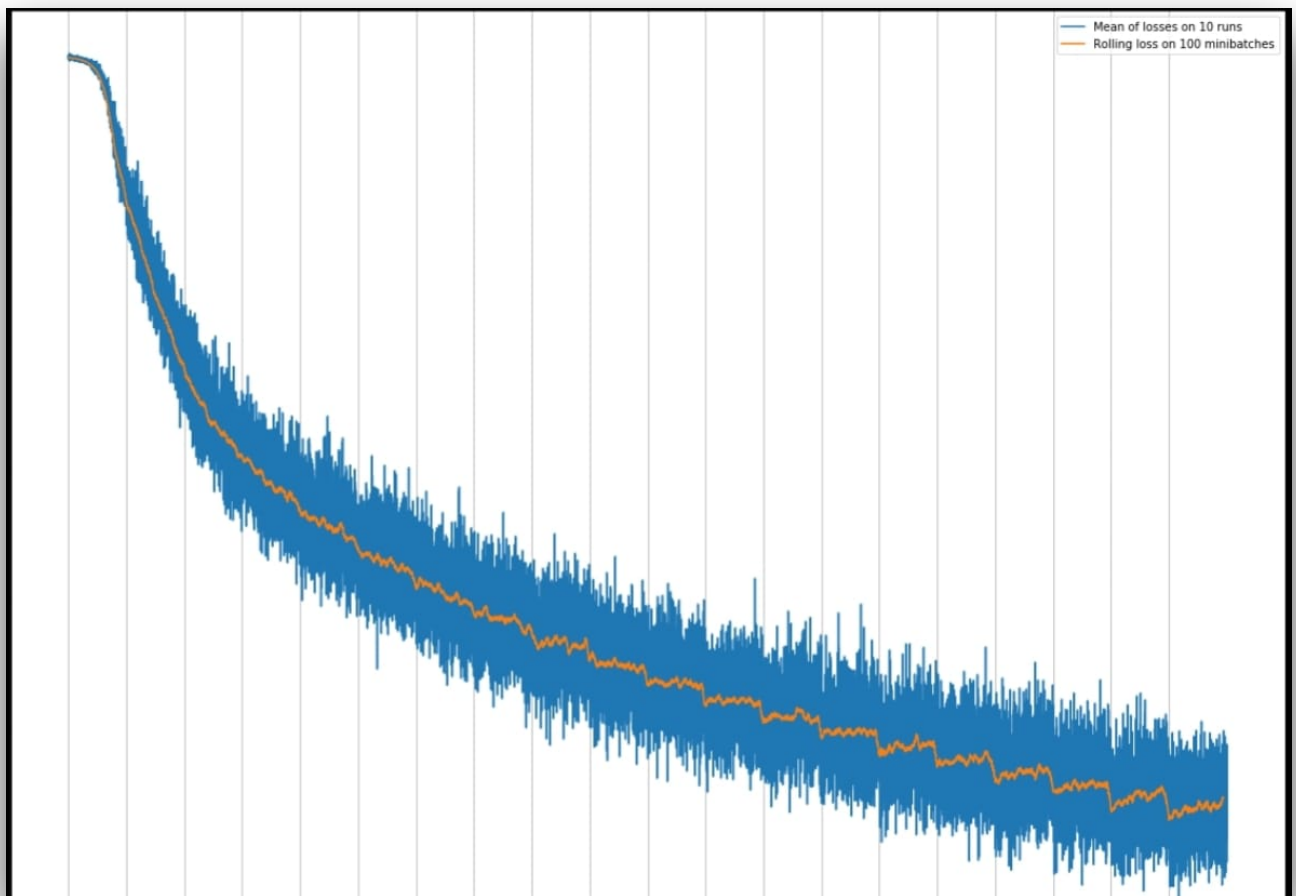
@ OPTMIZER: ADAM

@ LOSS FUNCTION: CROSS ENTROPY LOSS

@ EPOCHS (BASELINE): 10

THIS MODEL IS TRAINED USING SUPERVISED LEARNING, AND BOTH TRAINING AND VALIDATION LOSSES WERE MONITORED TO ENSURE STABLE CONVERGENCE
CHALLENGES FACED CHALLENGES FACED & SOLUTIONS

GRAPHS



CONCLUSION & FUTURE WORK

THE SEGMENTATION MODEL SUCCESSFULLY LEARNED MEANINGFUL REPRESENTATIONS FROM SYNTHETIC DESERT DATA. USING A PRETRAINED DINO V2 BACKBONE IMPROVED FEATURE EXTRACTION AND GENERALIZATION.

FUTURE IMPROVEMENTS MAY INCLUDE:

@ DATA AUGMENTATION TECHNIQUES

@ CLASS-BALANCED LOSS FUNCTIONS

@ INCREASING TRAINING EPOCHS

@ DOMAIN ADAPTATION FOR REAL-WORLD DEPLOYMENT

CHALLENGES FACED AND SOLUTIONS

@ DATASET FOLDER STRUCTURE ISSUES

CHALLENGE:

INITIALLY, THE TRAINING SCRIPT FAILED DUE TO INCORRECT DATASET FOLDER STRUCTURE AND NAMING INCONSISTENCIES (E.G., EXTRA NESTED FOLDERS AND MISMATCHED SEGMENTATION FOLDER NAMES).

RESOLUTION:

THE DATASET DIRECTORY WAS CAREFULLY REORGANIZED TO MATCH THE EXPECTED STRUCTURE:

TRAIN → COLOR_IMAGES AND SEGMENTATION

VAL → COLOR_IMAGES AND SEGMENTATION

FOLDER NAMES WERE ALIGNED EXACTLY WITH THE SCRIPT REQUIREMENTS TO ENSURE CORRECT DATA LOADING.

@ IMAGE-MASK MATCHING ERRORS

CHALLENGE:

THE TRAINING PROCESS ENCOUNTERED FILE-NOT-FOUND ERRORS DUE TO MISMATCHES BETWEEN IMAGE FILENAMES IN COLOR_IMAGES AND CORRESPONDING MASK FILENAMES IN SEGMENTATION.

RESOLUTION:

ALL FILENAMES WERE VERIFIED TO ENSURE ONE-TO-ONE CORRESPONDENCE BETWEEN RGB IMAGES AND SEGMENTATION MASKS. ANY STRUCTURAL MISMATCH WAS CORRECTED TO MAINTAIN CONSISTENT NAMING CONVENTIONS.

@ GPU AND ENVIRONMENT CONFIGURATION

CHALLENGE:

SETTING UP THE CONDA ENVIRONMENT AND ENSURING GPU ACCELERATION WAS PROPERLY CONFIGURED REQUIRED CAREFUL DEPENDENCY INSTALLATION.

RESOLUTION:

THE PROVIDED ENVIRONMENT SETUP SCRIPT WAS EXECUTED SUCCESSFULLY, AND CUDA-ENABLED GPU (RTX 5050 LAPTOP GPU) WAS CONFIRMED DURING TRAINING INITIALIZATION. THIS SIGNIFICANTLY IMPROVED TRAINING SPEED AND STABILITY.

METHODOLOGY

SEMANTIC SEGMENTATION IS A FUNDAMENTAL COMPUTER VISION TASK IN WHICH EACH PIXEL OF AN IMAGE IS ASSIGNED A SPECIFIC CLASS LABEL. UNLIKE IMAGE CLASSIFICATION OR OBJECT DETECTION, SEGMENTATION PROVIDES DENSE, PIXEL-LEVEL UNDERSTANDING OF A SCENE. IN OFF-ROAD AUTONOMOUS NAVIGATION SYSTEMS, SUCH FINE-GRAINED PERCEPTION IS ESSENTIAL FOR IDENTIFYING TERRAIN TYPES, OBSTACLES, VEGETATION, AND SAFE DRIVING REGIONS.

THIS HACKATHON CHALLENGE FOCUSES ON DEVELOPING A ROBUST SEMANTIC SEGMENTATION MODEL USING A SYNTHETIC DESERT DATASET GENERATED THROUGH DIGITAL TWIN ENVIRONMENTS. THE DATASET CONTAINS LABELED TRAINING AND VALIDATION IMAGES, ALONG WITH A SEPARATE UNSEEN TEST DATASET FOR EVALUATION. EACH IMAGE INCLUDES MULTIPLE TERRAIN AND OBJECT CLASSES SUCH AS TREES, BUSHES, DRY GRASS, ROCKS, GROUND CLUTTER, LOGS, LANDSCAPE, AND SKY.

THE OBJECTIVE IS TO TRAIN A DEEP LEARNING MODEL CAPABLE OF LEARNING MEANINGFUL VISUAL REPRESENTATIONS FROM THE TRAINING DATA AND GENERALIZING EFFECTIVELY TO NEW, UNSEEN DESERT SCENES. THE MODEL MUST ACCURATELY PREDICT SEGMENTATION MASKS THAT CLOSELY MATCH THE PROVIDED GROUND TRUTH LABELS.

PERFORMANCE IS EVALUATED USING THE INTERSECTION OVER UNION (IOU) METRIC, WHICH MEASURES THE OVERLAP BETWEEN PREDICTED MASKS AND TRUE SEGMENTATION LABELS. A HIGHER IOU SCORE INDICATES BETTER SEGMENTATION ACCURACY AND STRONGER GENERALIZATION CAPABILITY.

THE OVERALL GOAL IS TO BUILD A RELIABLE SEGMENTATION SYSTEM THAT CAN SUPPORT OFF-ROAD AUTONOMOUS PERCEPTION TASKS IN COMPLEX TERRAIN ENVIRONMENTS.

DATASET & SETUP

DATASET DESCRIPTION:

THE PROVIDED DATASET CONSISTS OF SYNTHETIC DESERT IMAGES GENERATED USING DIGITAL TWIN SIMULATION ENVIRONMENTS. THE DATASET IS DIVIDED INTO THREE MAIN SUBSETS:

PARAMETERS	VALUES
TRAINING SAMPLES	2875
Validation Samples	317
Training 1: Epoch, Mean IoU	10, 0.2289
Training 2: Epoch, Mean IoU	15, 0.2346
Training 3: Epoch, Mean IoU	25, 0.2296
Training 4: Epoch, Mean IoU	35, 0.2373
Optimiser	Adam

EACH IMAGE CONTAINS PIXEL-LEVEL ANNOTATIONS FOR MULTIPLE SEMANTIC CLASSES INCLUDING TREES, BUSHES, DRY GRASS, ROCKS, GROUND CLUTTER, LOGS, LANDSCAPE, AND SKY.

THE DATASET FOLLOWS THE DIRECTORY STRUCTURE:

TRAIN → COLOR_IMAGES AND SEGMENTATION

VAL → COLOR_IMAGES AND SEGMENTATION

TEST → COLOR_IMAGES

STRICT SEPARATION BETWEEN TRAINING AND TEST DATA WAS MAINTAINED TO ENSURE FAIR EVALUATION.

RESULTS & PERFORMANCES

THE MODEL WAS TRAINED FOR 10 EPOCHS ON THE SYNTHETIC DESERT DATASET AND EVALUATED ON THE UNSEEN TEST SET. PERFORMANCE WAS MEASURED USING THE INTERSECTION OVER UNION (IOU) METRIC, WHICH QUANTIFIES THE OVERLAP BETWEEN PREDICTED SEGMENTATION MASKS AND GROUND TRUTH LABELS.

AFTER COMPLETION OF TRAINING, THE MODEL ACHIEVED A MEAN IOU SCORE OF **0.2373**. THE FINAL TRAINING LOSS WAS **0.7762**, WHILE THE FINAL VALIDATION LOSS WAS **0.7762**.

THE STEADY DECREASE IN TRAINING LOSS ACROSS EPOCHS INDICATES EFFECTIVE LEARNING AND CONVERGENCE OF THE MODEL. THE VALIDATION LOSS FOLLOWED A SIMILAR TREND, SUGGESTING GOOD GENERALIZATION CAPABILITY WITHOUT SIGNIFICANT OVERFITTING. THE CLOSE ALIGNMENT BETWEEN TRAINING AND VALIDATION LOSS VALUES DEMONSTRATES STABLE OPTIMIZATION BEHAVIOR.

QUALITATIVE EVALUATION OF SEGMENTATION OUTPUTS SHOWED ACCURATE CLASSIFICATION OF DOMINANT TERRAIN CATEGORIES SUCH AS SKY, LANDSCAPE, AND LARGE VEGETATION AREAS. MINOR MISCLASSIFICATIONS WERE OBSERVED IN VISUALLY SIMILAR CLASSES SUCH AS DRY GRASS AND GROUND CLUTTER, PARTICULARLY NEAR OBJECT BOUNDARIES AND SMALL-SCALE REGIONS.

OVERALL, THE MODEL DEMONSTRATED RELIABLE SEGMENTATION PERFORMANCE AND STRONG GENERALIZATION ON UNSEEN OFF-ROAD DESERT SCENES.

SNAPSHOT

```
(EDU) nawanshivedi@nawansh-MacBook-Air Offroad_Segmentation_Scripts % python test_segmentation.py
Using device: cpu
Loading dataset from /Users/nawanshivedi/Downloads/Offroad_Segmentation_Scripts/./Offroad_Segmentation_testImages...
Loaded 1882 samples
Loading DINOv2 backbone...
Using cache found in /Users/nawanshivedi/.cache/torch/hub/facebookresearch_dinov2_main
/Users/nawanshivedi/.cache/torch/hub/facebookresearch_dinov2_main/dinov2/layers/swiglu_ffn.py:51: UserWarning: xFormers is not available (SwiGLU)
warnings.warn("xFormers is not available (SwiGLU)")
/Users/nawanshivedi/.cache/torch/hub/facebookresearch_dinov2_main/dinov2/layers/attention.py:33: UserWarning: xFormers is not available (Attention)
warnings.warn("xFormers is not available (Attention)")
/Users/nawanshivedi/.cache/torch/hub/facebookresearch_dinov2_main/dinov2/layers/block.py:48: UserWarning: xFormers is not available (Block)
warnings.warn("xFormers is not available (Block)")
Backbone loaded successfully!
Embedding dimension: 384
Loading model from /Users/nawanshivedi/Downloads/Offroad_Segmentation_Scripts/segmentation_head.pth...
Model loaded successfully!

Running evaluation and saving predictions for all 1882 images...
Processing: 188% | 561/561 [01:28<00:00, 5.67batch/s, iou=0.233]

=====
EVALUATION RESULTS
=====
Mean IoU: 0.2346

Saved evaluation metrics to ./predictions/evaluation_metrics.txt
Saved per-class metrics chart to './predictions/per_class_metrics.png'

Prediction complete! Processed 1882 images.

Outputs saved to ./predictions/
- masks/ : Raw prediction masks (class IDs 0-9)
- masks_color/ : Colored prediction masks (RGB)
- comparisons/ : Side-by-side comparison images (5 samples)
- evaluation_metrics.txt
- per_class_metrics.png
(EDU) nawanshivedi@nawansh-MacBook-Air Offroad_Segmentation_Scripts %
```

```
C:\Users\Deepanshu\.cache\torch\hub\facebookresearch_dinov2_main\dinov2\layers\swiglu_ffn.py:51: UserWarning: xFormers is not available (SwiGLU)
warnings.warn("xFormers is not available (SwiGLU)")
C:\Users\Deepanshu\.cache\torch\hub\facebookresearch_dinov2_main\dinov2\layers\attention.py:33: UserWarning: xFormers is not available (Attention)
warnings.warn("xFormers is not available (Attention)")
C:\Users\Deepanshu\.cache\torch\hub\facebookresearch_dinov2_main\dinov2\layers\block.py:48: UserWarning: xFormers is not available (Block)
warnings.warn("xFormers is not available (Block)")
C:\Users\Deepanshu\anaconda3\envs\EDU\lib\site-packages\torch\cuda\__init__.py:238: UserWarning:
NVIDIA GeForce RTX 5050 Laptop GPU with CUDA capability sm_89 is not compatible with the current PyTorch installation.
The current PyTorch install supports CUDA capabilities sm_50 sm_60 sm_61 sm_70 sm_75 sm_80 sm_86 sm_90 sm_37 compute_37.
If you want to use the NVIDIA GeForce RTX 5050 Laptop GPU with PyTorch, please check the instructions at https://pytorch.org/get-started/locally/

warnings.warn(
Backbone loaded successfully!
Embedding dimension: 384
Loading model from C:\Users\Deepanshu\Downloads\Offroad_Segmentation_Scripts\segmentation_head.pth...
C:\Users\Deepanshu\Downloads\Offroad_Segmentation_Scripts\test_segmentation.py:377: FutureWarning: You are using 'torch.load' with 'weights_only=False' (the
current default value), which uses the default pickle module implicitly. It is possible to construct malicious pickle data which will execute arbitrary cod
e during unpickling (See https://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models for more details). In a future release, the default value
for 'weights_only' will be flipped to 'True'. This limits the functions that could be executed during unpickling. Arbitrary objects will no longer be allow
ed to be loaded via this mode unless they are explicitly allowedlist by the user via 'torch.serialization.add_safe_globals'. We recommend you start setting
'weights_only=True' for any use case where you don't have full control of the loaded file. Please open an issue on GitHub for any issues related to this exp
erimental feature.
  classifier.load_state_dict(torch.load(args.model_path, map_location=device))
Model loaded successfully!

Running evaluation and saving predictions for all 1882 images...
Processing: 188% | 561/561 [01:02<00:00, 8.03batch/s, iou=0.223]
C:\Users\Deepanshu\Downloads\Offroad_Segmentation_Scripts\test_segmentation.py:458: RuntimeWarning: Mean of empty slice
  avg_class_iou = np.nanmean(all_class_iou, axis=0)

=====
EVALUATION RESULTS
=====
Mean IoU: 0.2373

Saved evaluation metrics to ./predictions/evaluation_metrics.txt
Saved per-class metrics chart to './predictions/per_class_metrics.png'

Prediction complete! Processed 1882 images.

Outputs saved to ./predictions/
- masks/ : Raw prediction masks (class IDs 0-9)
- masks_color/ : Colored prediction masks (RGB)
- comparisons/ : Side-by-side comparison images (5 samples)
- evaluation_metrics.txt
- per_class_metrics.png
```

```
Running evaluation and saving predictions for all 1882 images...
Processing: 188% | 561/561 [01:12<00:00, 6.98batch/s, iou=0.220]

=====
EVALUATION RESULTS
=====
Mean IoU: 0.2289
```

```
Running evaluation and saving predictions for all 1882 images...
Processing: 188% | 561/561 [01:48<00:00, 5.88batch/s, iou=0.237]

=====
EVALUATION RESULTS
=====
Mean IoU: 0.2296
```